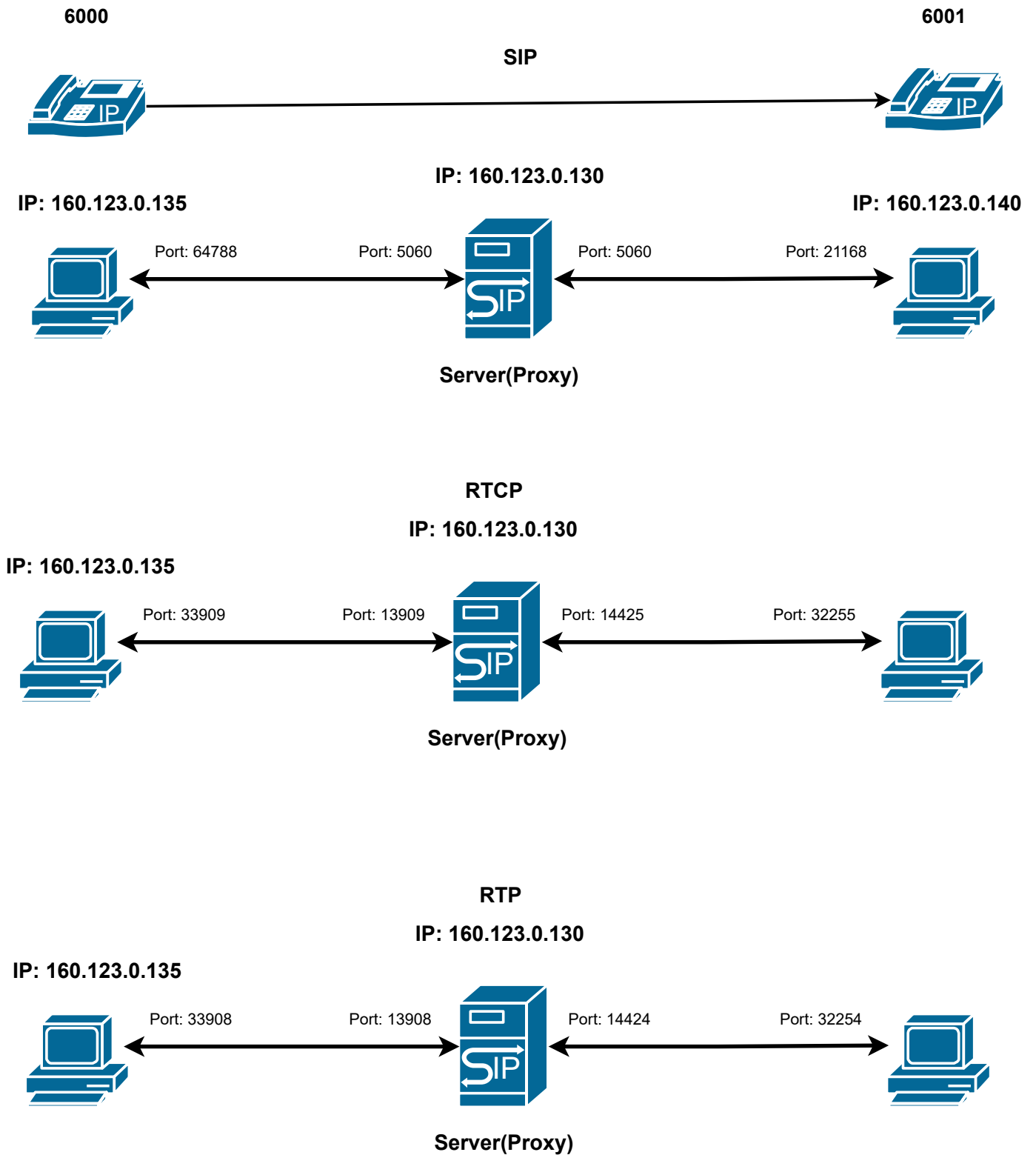




**VOIP  
COS 370**

**الاسم: مالك سالم العنزي  
الرقم: 443170256**

# Topology

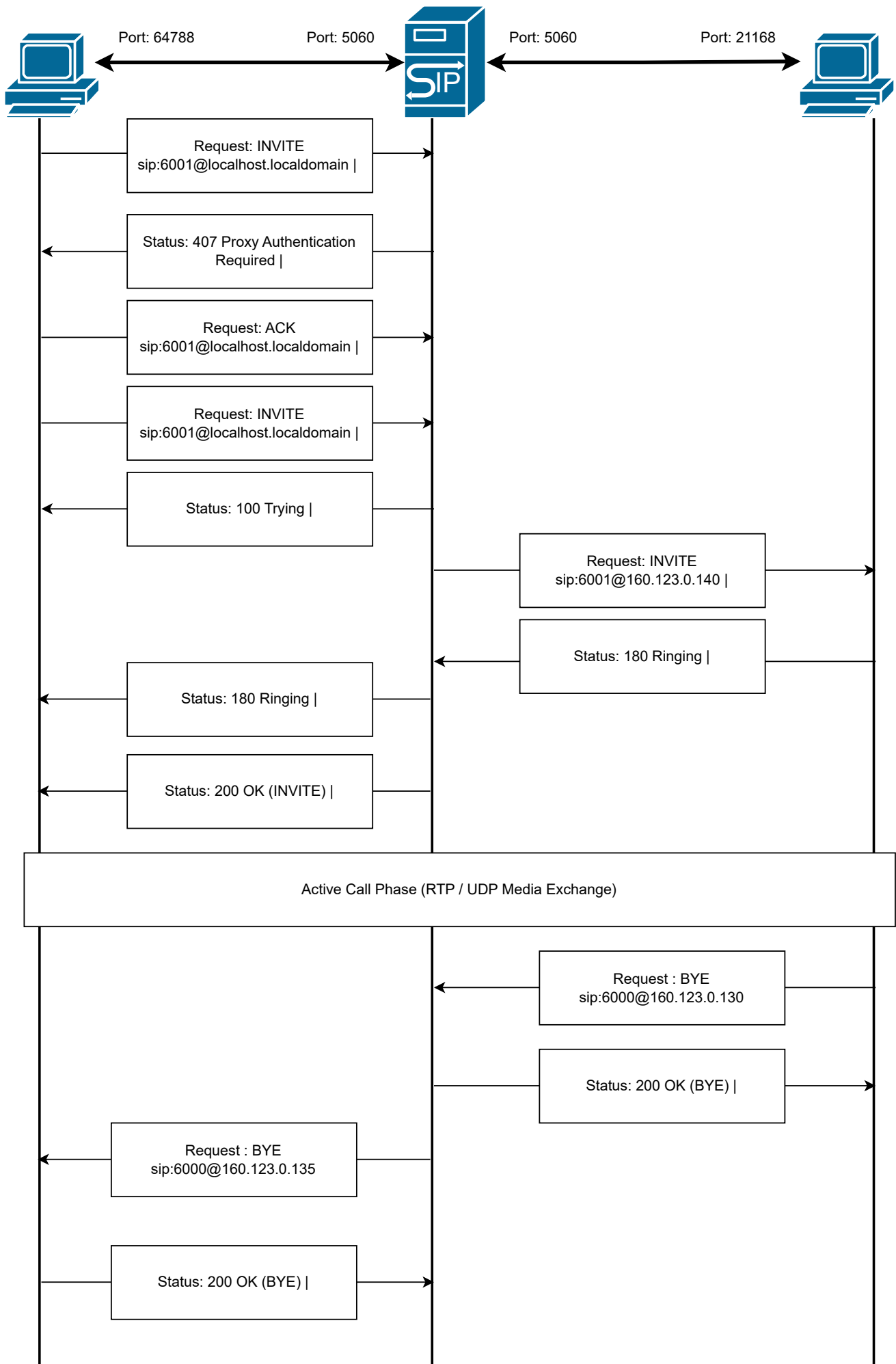


| P#            | client-1                              | Server                                                                | client-2                             |
|---------------|---------------------------------------|-----------------------------------------------------------------------|--------------------------------------|
| Packet number | IP: 160.123.0.135                     | IP: 160.123.0.130                                                     | IP: 160.123.0.140                    |
| 48            | INVITE sip:6001@localhost.localdomain |                                                                       |                                      |
| 49            |                                       | 407 Proxy Authentication Required                                     |                                      |
| 50            | ACK sip:6001@localhost.localdomain    |                                                                       |                                      |
| 51            | INVITE sip:6001@localhost.localdomain |                                                                       |                                      |
| 52            |                                       | Status: 100 Trying                                                    |                                      |
| 53            |                                       | INVITE<br>sip:6001@160.123.0.140:21168;rinstance=<br>1dab1ccbf8fe5755 |                                      |
| 54            |                                       |                                                                       | Status: 180 Ringing                  |
| 55            |                                       | Status: 180 Ringing                                                   |                                      |
| 63            |                                       | Status:200 OK (INVITE)                                                |                                      |
| 1646          |                                       |                                                                       | Request : BYE sip:6000@160.123.0.130 |
| 1647          |                                       | Status:200 OK (BYE)                                                   |                                      |
| 1648          |                                       | Request : BYE sip:6000@160.123.0.135                                  |                                      |
| 1650          | Status:200 OK (BYE)                   |                                                                       |                                      |

IP: 160.123.0.135

IP: 160.123.0.130

IP: 160.123.0.140



## RTCP from client-2 to Server

|                   |                |          |           |                    |
|-------------------|----------------|----------|-----------|--------------------|
| Version: 2        | Padding: False | Count: 0 | Type: 201 | Length: 1(8 bytes) |
| Sender Info Block |                |          |           |                    |
| Report Block      |                |          |           |                    |

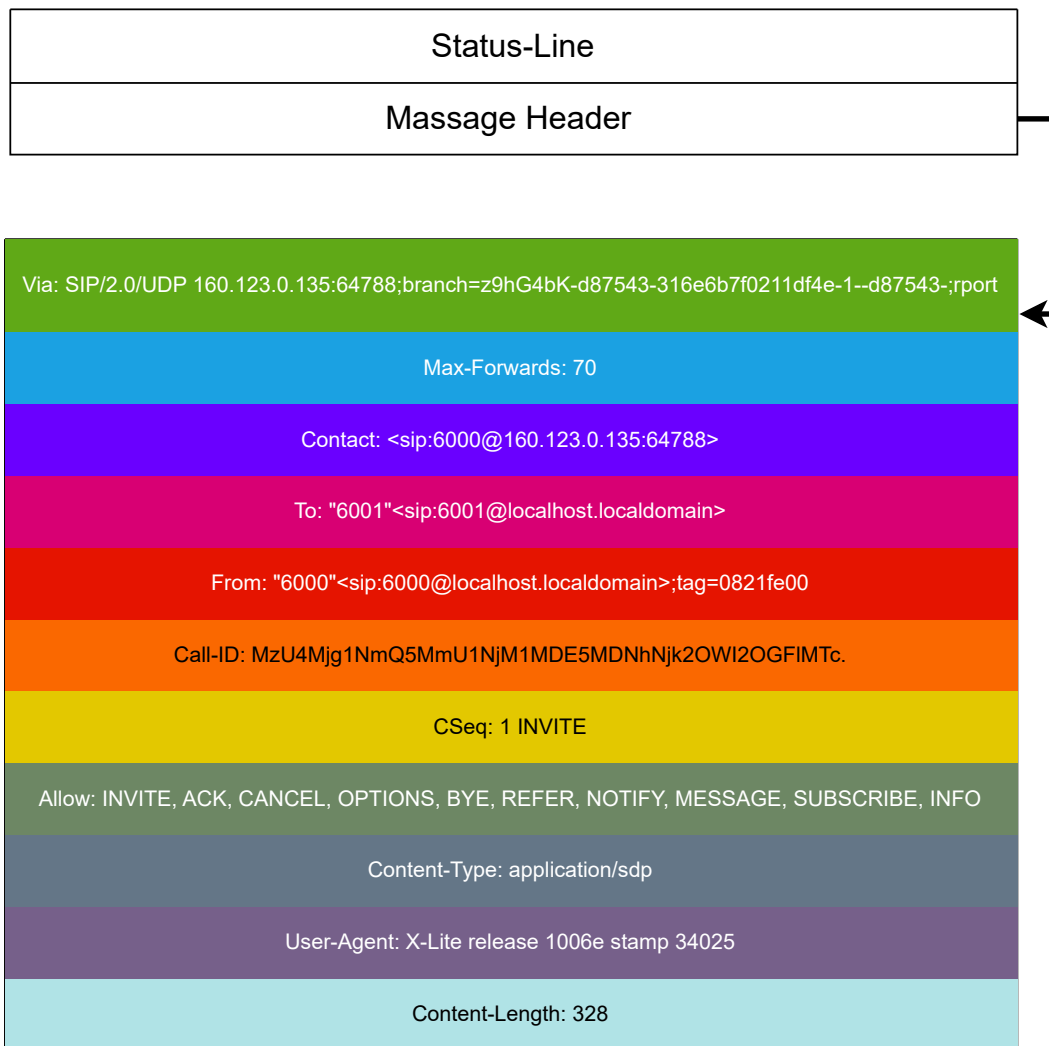
|      |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |
|------|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|
| 0000 | 00 | 0c | 29 | ef | e1 | ee | 00 | 0c | 29 | a4 | 39 | be | 08 | 00 | 45 | 00 |
| 0010 | 00 | a0 | 15 | 86 | 00 | 00 | 80 | 11 | e2 | c2 | a0 | 7b | 00 | 8c | a0 | 7b |
| 0020 | 00 | 82 | 7d | ff | 38 | 59 | 00 | 8c | 44 | d4 | 80 | c9 | 00 | 01 | 72 | 2f |
| 0030 | a3 | 17 | 81 | ca | 00 | 1e | 72 | 2f | a3 | 17 | 01 | 3d | 43 | 35 | 30 | 41 |
| 0040 | 36 | 41 | 34 | 35 | 31 | 38 | 38 | 46 | 34 | 43 | 37 | 44 | 38 | 35 | 44 | 46 |
| 0050 | 45 | 35 | 34 | 36 | 36 | 42 | 30 | 33 | 35 | 46 | 42 | 45 | 40 | 75 | 6e | 69 |
| 0060 | 71 | 75 | 65 | 2e | 7a | 31 | 37 | 36 | 43 | 31 | 32 | 30 | 30 | 45 | 46 | 30 |
| 0070 | 45 | 34 | 36 | 34 | 37 | 2e | 6f | 72 | 67 | 08 | 31 | 10 | 78 | 2d | 72 | 74 |
| 0080 | 70 | 2d | 73 | 65 | 73 | 73 | 69 | 6f | 6e | 2d | 69 | 64 | 39 | 44 | 42 | 45 |
| 0090 | 32 | 37 | 32 | 34 | 46 | 42 | 39 | 44 | 34 | 34 | 42 | 46 | 39 | 32 | 44 | 33 |
| 00a0 | 35 | 33 | 41 | 44 | 45 | 41 | 35 | 32 | 33 | 44 | 35 | 35 | 00 | 00 |    |    |

## RTCP from client-1 to Server

|                   |                |          |           |                    |
|-------------------|----------------|----------|-----------|--------------------|
| Version: 2        | Padding: False | Count: 0 | Type: 201 | Length: 1(8 bytes) |
| Sender Info Block |                |          |           |                    |
| Report Block      |                |          |           |                    |

|      |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |
|------|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|
| 0000 | 00 | 0c | 29 | ef | e1 | ee | 00 | 0c | 29 | 9c | c3 | bd | 08 | 00 | 45 | 00 |
| 0010 | 00 | a0 | 15 | 3d | 00 | 00 | 80 | 11 | e3 | 10 | a0 | 7b | 00 | 87 | a0 | 7b |
| 0020 | 00 | 82 | 84 | 75 | 36 | 55 | 00 | 8c | 0d | 8b | 80 | c9 | 00 | 01 | 4b | cd |
| 0030 | 70 | 73 | 81 | ca | 00 | 1e | 4b | cd | 70 | 73 | 01 | 3d | 34 | 44 | 31 | 35 |
| 0040 | 41 | 37 | 38 | 33 | 30 | 36 | 32 | 39 | 34 | 36 | 39 | 35 | 41 | 33 | 31 | 45 |
| 0050 | 34 | 43 | 43 | 36 | 34 | 43 | 38 | 35 | 41 | 32 | 37 | 42 | 40 | 75 | 6e | 69 |
| 0060 | 71 | 75 | 65 | 2e | 7a | 46 | 32 | 37 | 45 | 45 | 35 | 34 | 43 | 43 | 33 | 38 |
| 0070 | 41 | 34 | 39 | 35 | 31 | 2e | 6f | 72 | 67 | 08 | 31 | 10 | 78 | 2d | 72 | 74 |
| 0080 | 70 | 2d | 73 | 65 | 73 | 73 | 69 | 6f | 6e | 2d | 69 | 64 | 38 | 46 | 35 | 36 |
| 0090 | 30 | 36 | 38 | 41 | 33 | 30 | 42 | 44 | 34 | 32 | 42 | 44 | 42 | 39 | 35 | 46 |
| 00a0 | 41 | 33 | 38 | 34 | 44 | 45 | 34 | 30 | 32 | 31 | 44 | 44 | 00 | 00 |    |    |

## SIP Message from client-1 to server



**From Wireshark frame number 48**

|      |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |
|------|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|
| 0000 | 00 | 0c | 29 | ef | e1 | ee | 00 | 0c | 29 | 9c | c3 | bd | 08 | 00 | 45 | 00 |
| 0010 | 03 | 91 | 15 | 3a | 40 | 00 | 80 | 11 | a0 | 22 | a0 | 7b | 00 | 87 | a0 | 7b |
| 0020 | 00 | 82 | fd | 14 | 13 | c4 | 03 | 7d | bd | 0f | 49 | 4e | 56 | 49 | 54 | 45 |
| 0030 | 20 | 73 | 69 | 70 | 3a | 36 | 30 | 30 | 31 | 40 | 6c | 6f | 63 | 61 | 6c | 68 |
| 0040 | 6f | 73 | 74 | 2e | 6c | 6f | 63 | 61 | 6c | 64 | 6f | 6d | 61 | 69 | 6e | 20 |
| 0050 | 53 | 49 | 50 | 2f | 32 | 2e | 30 | 0d | 0a | 56 | 69 | 61 | 3a | 20 | 53 | 49 |
| 0060 | 50 | 2f | 32 | 2e | 30 | 2f | 55 | 44 | 50 | 20 | 31 | 36 | 30 | 2e | 31 | 32 |
| 0070 | 33 | 2e | 30 | 2e | 31 | 33 | 35 | 3a | 36 | 34 | 37 | 38 | 38 | 3b | 62 | 72 |
| 0080 | 61 | 6e | 63 | 68 | 3d | 7a | 39 | 68 | 47 | 34 | 62 | 4b | 2d | 64 | 38 | 37 |
| 0090 | 35 | 34 | 33 | 2d | 33 | 31 | 36 | 65 | 36 | 62 | 37 | 66 | 30 | 32 | 31 | 31 |
| 00a0 | 64 | 66 | 34 | 65 | 2d | 31 | 2d | 2d | 64 | 38 | 37 | 35 | 34 | 33 | 2d | 3b |
| 00b0 | 72 | 70 | 6f | 72 | 74 | 0d | 0a | 4d | 61 | 78 | 2d | 46 | 6f | 72 | 77 | 61 |
| 00c0 | 72 | 64 | 73 | 3a | 20 | 37 | 30 | 0d | 0a | 43 | 6f | 6e | 74 | 61 | 63 | 74 |
| 00d0 | 3a | 20 | 3c | 73 | 69 | 70 | 3a | 36 | 30 | 30 | 30 | 40 | 31 | 36 | 30 | 2e |
| 00e0 | 31 | 32 | 33 | 2e | 30 | 2e | 31 | 33 | 35 | 3a | 36 | 34 | 37 | 38 | 38 | 3e |
| 00f0 | 0d | 0a | 54 | 6f | 3a | 20 | 22 | 36 | 30 | 30 | 31 | 22 | 3c | 73 | 69 | 70 |
| 0100 | 3a | 36 | 30 | 30 | 31 | 40 | 6c | 6f | 63 | 61 | 6c | 68 | 6f | 73 | 74 | 2e |
| 0110 | 6c | 6f | 63 | 61 | 6c | 64 | 6f | 6d | 61 | 69 | 6e | 3e | 0d | 0a | 46 | 72 |
| 0120 | 6f | 6d | 3a | 20 | 22 | 36 | 30 | 30 | 30 | 22 | 3c | 73 | 69 | 70 | 3a | 36 |
| 0130 | 30 | 30 | 30 | 40 | 6c | 6f | 63 | 61 | 6c | 68 | 6f | 73 | 74 | 2e | 6c | 6f |
| 0140 | 63 | 61 | 6c | 64 | 6f | 6d | 61 | 69 | 6e | 3e | 3b | 74 | 61 | 67 | 3d | 30 |
| 0150 | 38 | 32 | 31 | 66 | 65 | 30 | 30 | 0d | 0a | 43 | 61 | 6c | 6c | 2d | 49 | 44 |
| 0160 | 3a | 20 | 4d | 7a | 55 | 34 | 4d | 6a | 67 | 31 | 4e | 6d | 51 | 35 | 4d | 6d |
| 0170 | 55 | 31 | 4e | 6a | 4d | 31 | 4d | 44 | 45 | 35 | 4d | 44 | 4e | 68 | 4e | 6a |
| 0180 | 6b | 32 | 4f | 57 | 49 | 32 | 4f | 47 | 46 | 6c | 4d | 54 | 63 | 2e | 0d | 0a |
| 0190 | 43 | 53 | 65 | 71 | 3a | 20 | 31 | 20 | 49 | 4e | 56 | 49 | 54 | 45 | 0d | 0a |
| 01a0 | 41 | 6c | 6c | 6f | 77 | 3a | 20 | 49 | 4e | 56 | 49 | 54 | 45 | 2c | 20 | 41 |
| 01b0 | 43 | 4b | 2c | 20 | 43 | 41 | 4e | 43 | 45 | 4c | 2c | 20 | 4f | 50 | 54 | 49 |
| 01c0 | 4f | 4e | 53 | 2c | 20 | 42 | 59 | 45 | 2c | 20 | 52 | 45 | 46 | 45 | 52 | 2c |
| 01d0 | 20 | 4e | 4f | 54 | 49 | 46 | 59 | 2c | 20 | 4d | 45 | 53 | 53 | 41 | 47 | 45 |
| 01e0 | 2c | 20 | 53 | 55 | 42 | 53 | 43 | 52 | 49 | 42 | 45 | 2c | 20 | 49 | 4e | 46 |
| 01f0 | 4f | 0d | 0a | 43 | 6f | 6e | 74 | 65 | 6e | 74 | 2d | 54 | 79 | 70 | 65 | 3a |
| 0200 | 20 | 61 | 70 | 70 | 6c | 69 | 63 | 61 | 74 | 69 | 6f | 6e | 2f | 73 | 64 | 70 |
| 0210 | 0d | 0a | 55 | 73 | 65 | 72 | 2d | 41 | 67 | 65 | 6e | 74 | 3a | 20 | 58 | 2d |
| 0220 | 4c | 69 | 74 | 65 | 20 | 72 | 65 | 6c | 65 | 61 | 73 | 65 | 20 | 31 | 30 | 30 |
| 0230 | 36 | 65 | 20 | 73 | 74 | 61 | 6d | 70 | 20 | 33 | 34 | 30 | 32 | 35 | 0d | 0a |
| 0240 | 43 | 6f | 6e | 74 | 65 | 6e | 74 | 2d | 4c | 65 | 6e | 67 | 74 | 68 | 3a | 20 |
| 0250 | 33 | 32 | 38 | 0d | 0a | 0d | 0a | 76 | 3d | 30 | 0d | 0a | 6f | 3d | 2d | 20 |
| 0260 | 32 | 20 | 32 | 20 | 49 | 4e | 20 | 49 | 50 | 34 | 20 | 31 | 36 | 30 | 2e | 31 |
| 0270 | 32 | 33 | 2e | 30 | 2e | 31 | 33 | 35 | 0d | 0a | 73 | 3d | 43 | 6f | 75 | 6e |
| 0280 | 74 | 65 | 72 | 50 | 61 | 74 | 68 | 20 | 58 | 2d | 4c | 69 | 74 | 65 | 20 | 33 |
| 0290 | 2e | 30 | 0d | 0a | 63 | 3d | 49 | 4e | 20 | 49 | 50 | 34 | 20 | 31 | 36 | 30 |
| 02a0 | 2e | 31 | 32 | 33 | 2e | 30 | 2e | 31 | 33 | 35 | 0d | 0a | 74 | 3d | 30 | 20 |
| 02b0 | 30 | 0d | 0a | 6d | 3d | 61 | 75 | 64 | 69 | 6f | 20 | 33 | 33 | 39 | 30 | 38 |
| 02c0 | 20 | 52 | 54 | 50 | 2f | 41 | 56 | 50 | 20 | 31 | 30 | 37 | 20 | 31 | 31 | 39 |
| 02d0 | 20 | 30 | 20 | 39 | 38 | 20 | 38 | 20 | 33 | 20 | 31 | 30 | 31 | 0d | 0a | 61 |
| 02e0 | 3d | 61 | 6c | 74 | 3a | 31 | 20 | 31 | 20 | 3a | 20 | 74 | 54 | 44 | 4a | 4e |
| 02f0 | 39 | 79 | 58 | 20 | 76 | 78 | 6e | 37 | 6e | 36 | 36 | 78 | 20 | 31 | 36 | 30 |
| 0300 | 2e | 31 | 32 | 33 | 2e | 30 | 2e | 31 | 33 | 35 | 20 | 33 | 33 | 39 | 30 | 38 |
| 0310 | 0d | 0a | 61 | 3d | 66 | 6d | 74 | 70 | 3a | 31 | 30 | 31 | 20 | 30 | 2d | 31 |
| 0320 | 35 | 0d | 0a | 61 | 3d | 72 | 74 | 70 | 6d | 61 | 70 | 3a | 31 | 30 | 37 | 20 |
| 0330 | 42 | 56 | 33 | 32 | 2f | 31 | 36 | 30 | 30 | 30 | 0d | 0a | 61 | 3d | 72 | 74 |
| 0340 | 70 | 6d | 61 | 70 | 3a | 31 | 31 | 39 | 20 | 42 | 56 | 33 | 32 | 2d | 46 | 45 |
| 0350 | 43 | 2f | 31 | 36 | 30 | 30 | 30 | 0d | 0a | 61 | 3d | 72 | 74 | 70 | 6d | 61 |
| 0360 | 70 | 3a | 39 | 38 | 20 | 69 | 4c | 42 | 43 | 2f | 38 | 30 | 30 | 30 | 0d | 0a |
| 0370 | 61 | 3d | 72 | 74 | 70 | 6d | 61 | 70 | 3a | 31 | 30 | 31 | 20 | 74 | 65 | 6c |
| 0380 | 65 | 70 | 68 | 6f | 6e | 65 | 2d | 65 | 76 | 65 | 6e | 74 | 2f | 38 | 30 | 30 |
| 0390 | 30 | 0d | 0a | 61 | 3d | 73 | 65 | 6e | 64 | 72 | 65 | 63 | 76 | 0d | 0a |    |